MILITARY SPECIFICATION SHEET

PLASTIC SHEET, LAMINATED, METAL-CLAD (FOR PRINTED WIRING BOARDS), BASE MATERIAL PX, PAPER BASE, EPOXY RESIN, FLAME RESISTANT, COPPER-CLAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the plastic sheets described herein shall consist of this specification and the latest issue of MIL-P-13949.

REQUIREMENTS:

Property values: See table I.

TABLE I. Property values.

Property to be tested	Test condition 1/	Thickness	Values required
1. Peel strength, minimum in lbs-per-inch width			
A. 1-oz/ft^2 copper			
As received	A	All	8
After solder float	A	All	8
At elevated temperature	105°C	All	5
After exposure to processing solutions	N/A	N/A	2/
B. 2-oz/ft^2 copper			
As received	A	All	10
After solder float	A	All	10
At elevated temperature	105°C	All	6
After exposure to processing solutions	N/A	N/A	2/
C. 3-, 4-, and 5-oz/ft^2 copper			
As received	A	All	12
After solder float	A	All	12
At elevated temperature	105°C	All	7
After exposure to processing solutions	N/A	N/A	2/
2. Volume resistivity, minimum in megohm-centimeter			
After moisture resistance	F	All	10^5
At elevated temperature	E-24/105	All	10^5
3. Surface resistivity, minimum megohm			
After moisture resistance	F	All	10^4
At elevated temperature	E-24/105	All	10^3

See footnotes at end of table.

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TABLE I. Property values - Continued.

Property to be tested	Test condition 1/	Thickness	Values required
		(inch)	
4. Water absorption, maximum percent	E-1/105+des +D-24/23	0.031	1.00
		0.062	0.65
		0.093	0.60
		0.125	0.50
		0.250	0.25
5. Dielectric breakdown (parallel to laminations), minimum kV, step-by-step	D-48/50+ D-0.5/23	All	30
6. Permittivity at 1 MHz max	D-24/23	All	4.8
at 50 MHz max	D-24/23	All	4.8
7. Loss tangent at 1 MHz max	D-24/23	All	0.040
at 50 MHz max	D-24/23	All	0.040
8. Q (resonance), minimum	D-24/23		
at 1 MHz		All	30
at 50 MHz		All	30
at 100 MHz		All	30
9. Flexural strength, minimum thousand lbs-per-square inch			
A. Lengthwise	A	All	20
B. Crosswise	A	All	15
C. Lengthwise at elevated temperature	N/A	N/A	N/A
10. Arc resistance, minimum seconds	D-48/50+ D-0.5/23	All	60
11. Flammability, maximum burntime seconds	A	All	15

1/ See specimen conditioning requirements, conditioning tolerances and testing information table XXVI of MIL-P-13949.

2/ No values are included for this type because this conditioning does not simulate processes currently in use for plating on this grade and therefore is not applicable.

Marking: Internal marking shall be red in color (when specified).

Resistance to solvents: Not applicable.

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NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
NSA - S5
DLA - ES

User activities:
Army - ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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